10.6 Elements of automation

1. Source of power
2. Feedback control
3. Machine programming

ROBOTICS

10.7 Introduction

Robotics is an interdisciplinary branch of engineering and science that includes mechanical engineering, electronics engineering, computer science, and others. Many definitions have been suggested for what we call a robot. (The word may conjure up various level of technological specification, ranging from a simple material handling device to a humanoid). It is widely accepted that today's robots used in industries originated in the invention of programmed material handling device by George. C. Devil in 1954.

S2B5BV

FIRST ROBOT IN THE WORLD

உங்களுக்குத் தெரியுமா?

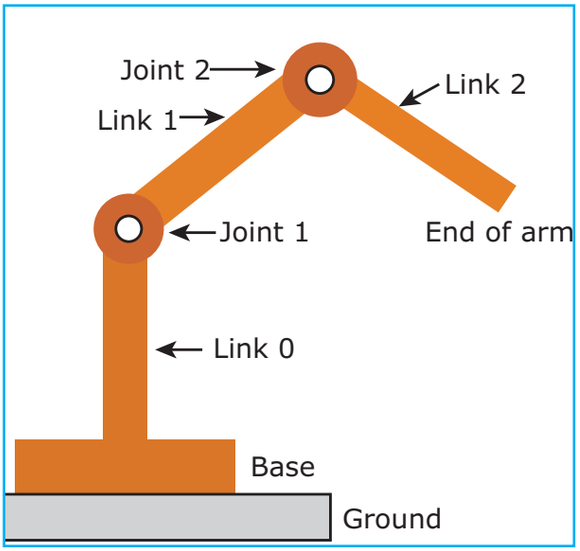
Father of Robotics Mr. George Devol

In 1954 George Devil invented the first digitally operated and a programmable robot called the Unimate. In 1956, Devil and his partner Joseph Engelberger formed the world's first robot company. In 1961, the first industrial robot, Unimate, went online in a General Motors automobile factory in New Jersey. Oct 16, 2017.





10.8 Definition-Robotics



The latest technologies are used to develop machines that can substitute for humans and replicate human actions are called robot.

The robots are mainly used in dangerous environments, manufacturing process or where human cannot survive.

Autonomy

Interaction between the human control and machine motions.

1. Tele operation:
   A human controls each movement, each machine actuator change is specified by the operator.
2. Supervisory:
   A human specifies general moves or position changes and the machine decides specific movements of its actuators.
3. Task – Level – autonomy
   The operator specifies only the task and the robot manage itself to compose it.
4. Fully autonomy
   The machine will create and complete all its task without human interaction

XII BME_English version CHAPTER 10.indd 171 1/8/2020 1:37:58 PM